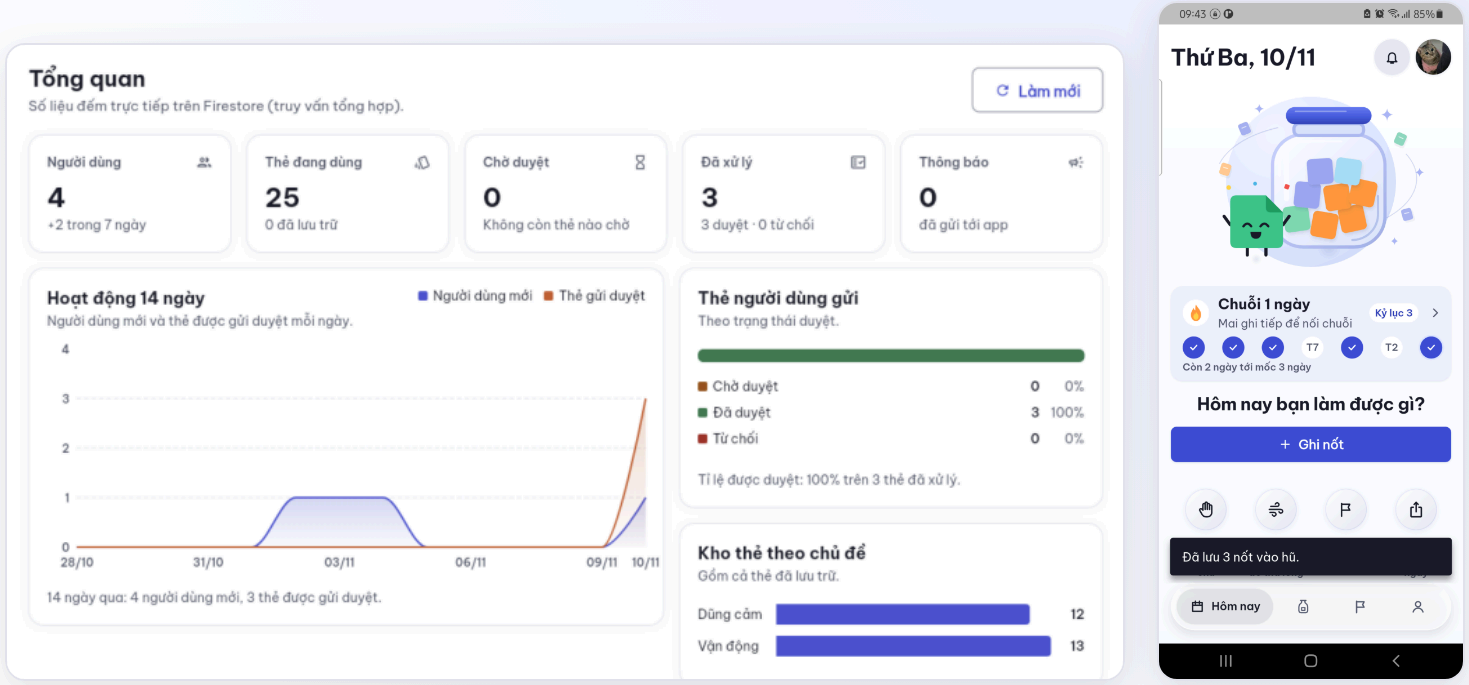
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Nốt

One positive note every day, now with staff, AI and a shared catalog

Đoàn Thị Kim Thúy · SE190992
Phùng Thế Quân · SE193519
Phan Đức Duy · SE193591
Phan Hữu Phước · SE161951



AGENDA FOUR PARTS

Problem → Design → Demo → Proof

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01

Problem

Why build it, and what is it?

Why build it?

- 01 Overview
- 02 Context
- 03 Problem

What is it?

- 04 Solution
- 05 Actors
- 06 Features

02

Design

How is the whole system built?

- 07 Business flow
- 08 State machine
- 09 Architecture
- 10 Stack
- 11 UI/UX
- 12 Sequence
- 13 Classes
- 14–15 Database
- 16 Traceability

03

Demo

How do the workflows run for real?

- 17 Demo plan
- 18 WF-01 Nốt AI
- 19 WF-02 Publish a card
- 20 WF-03 Announce

04

Proof

What can we prove?

- 21 Testing
- 22 Achievements
- 23 Limitations
- 24 Conclusion

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01

Problem

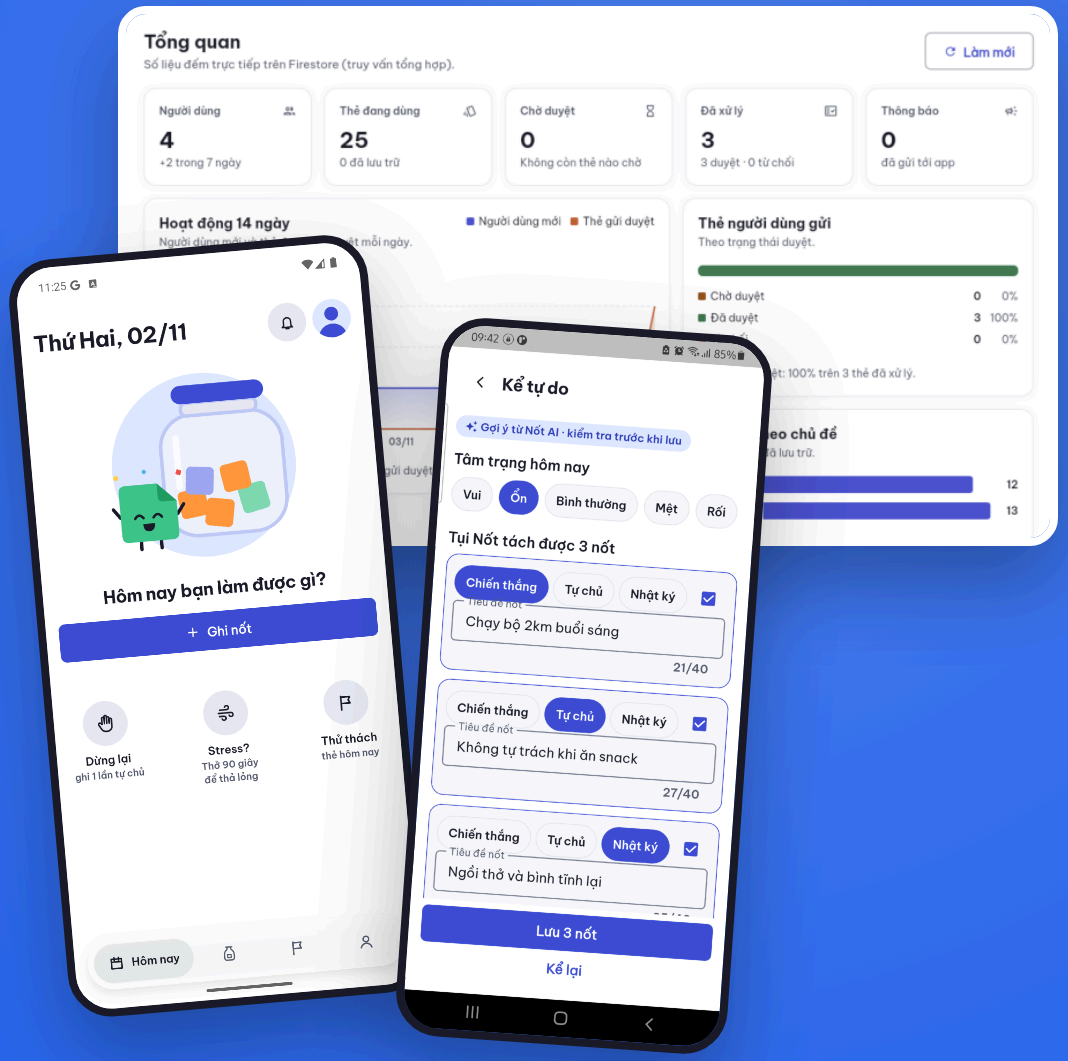
Why build it, and what is it?

Why build it?

01 Overview | 02 Context | 03 Problem

What is it?

04 Solution | 05 Actors | 06 Features



Nốt at a glance

PROJECT

Nốt

Small-wins journal · Android app + web admin · one Firebase project

> For **students and young workers** who end a long day feeling they got nothing done: Nốt makes **one small win visible in under 30 seconds** and keeps **fresh small challenges** coming.

USERS

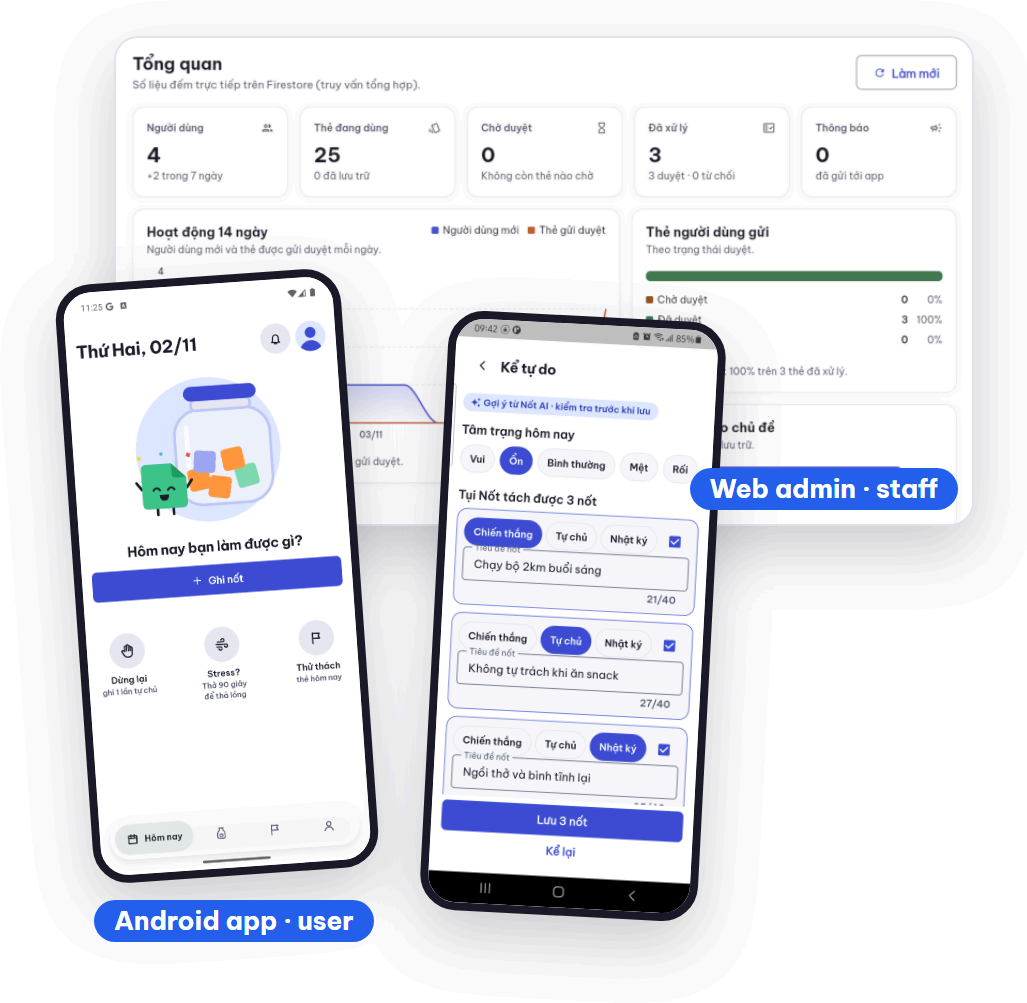
user
logs the day on the Android app

staff
reviews and creates cards, posts news on the web admin

admin
everything staff can, plus changes roles

CORE VALUE

- One win in 30 s, typed or sorted by Gemini
- New challenge cards without an app update
- Staff news in every inbox in the same minute



Context: an evening without Nốt

Final-year students and young workers reach the evening with little energy left.

ENVIRONMENT

University life and first jobs

long days of classes, internships, part-time work; the evening is the only free time

WHO IS INVOLVED



Minh, 22 · final-year student, tired, wants to feel the day counted

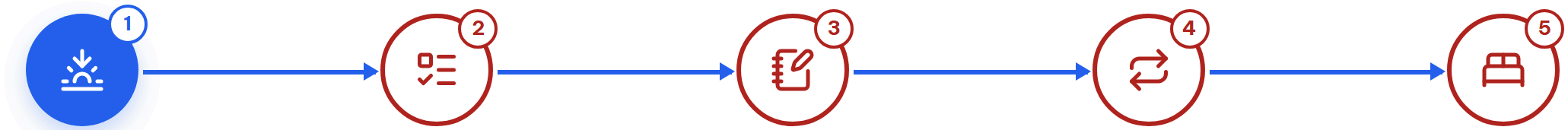
Nobody curates small, fresh challenges for people like Minh

CURRENT TOOLS

To-do app, phone notes, paper diary

plus self-help tips scattered across social media

CURRENT PROCESS



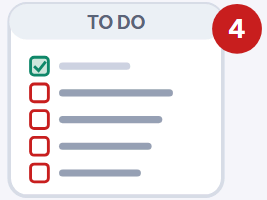
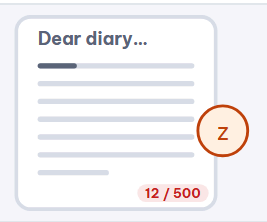
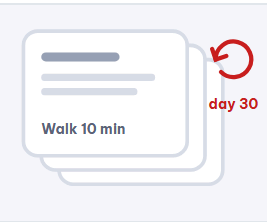
1. **Day ends** — home late, low energy
2. **Checks to-do** — sees only what is left
3. **Tries a diary** — blank page, gives up
4. **Same goals** — last week's list again
5. **Goes to bed** — "nothing done today"

So people need: a way to see one small win in seconds, and fresh small challenges that someone keeps up for them.

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“I didn’t get anything done today”

Most of us think it after a long day, even on days when we did plenty of small things.

PAIN POINT		CAUSE		IMPACT
	**At night we only see what is unfinished**	→	To-do lists track **what is left**; the small things we did are never written down	→ We feel the day was wasted; motivation drops and **burnout** builds up
	**Writing the day down is one more task**	→	Diary apps ask for a **long entry**, at the moment we are most tired	→ We skip it, then **stop after a few days**; the good moments are lost
	**Goals that never change stop motivating**	→	**Nobody refreshes** the challenges or tells us what is new	→ The same list every week gets boring; good habits **break off**

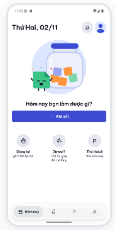
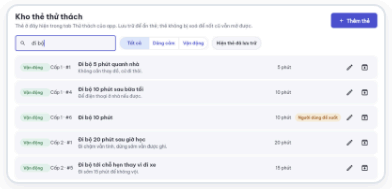
HIT HARDEST final-year students and young workers, whose evening is the only free time and the most tired one.

EVIDENCE Lab 2 persona (Minh, 22) and our own evenings as final-year students. No formal survey was run.

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Proposed solution: one Firebase system

An Android app for the evening, a staff web admin for content, and Gemini that sorts the day.

PAIN POINT		SOLUTION IN NỐT		VALUE
At night we only see what is unfinished	→	 **The jar of small wins**: one mood + one win, saved in under 30 s	→	The evening ends on **what we did**, not on what is left
Writing the day down is one more task	→	 **Nốt AI**: tell the day in one paragraph; Gemini splits it into sorted notes to confirm	→	**3 notes** from one paragraph, no form to fill
Goals that never change stop motivating	→	 **Staff web admin**: keep the card catalog fresh, approve cards users share, announce news	→	New cards and news reach users **without an app update**

IN SCOPE WF-01 Log the day with Nốt AI · WF-02 Publish a user card · WF-03 Announce news · onboarding, weekly PDF, roles, App Check

OUT OF SCOPE Cloud Functions and Cloud Storage (paid plan) · "Hỏi Nốt" AI (bonus) · iOS build

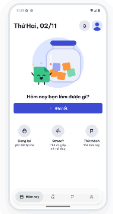
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Three roles, two external services

The role lives in users/{uid}.role; only an admin changes it, and never their own.

User
logs every evening

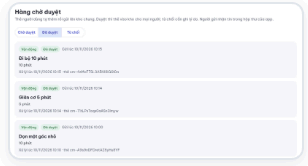
- own notes and data
- share own cards (pending only)
- read catalog, news, own inbox



Android app

Staff
content curator

- create, edit, archive cards
- approve / reject with a reason
- post announcements



Web admin

Admin
all staff rights
team lead

- everything staff can do
- change other users' roles
- never their own role



Web admin

Firebase
Auth · Firestore
App Check · Remote Config · FCM

Security Rules decide every read and write

Gemini
via Firebase AI Logic
gets the scrubbed paragraph only
returns JSON suggestions

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Three core workflows

Each workflow has an actor, a trigger and a business result, not a list of buttons.

WF-01

Log the day with Nốt AI

ACTOR	User
TRIGGER	Evening, user taps Ghi nốt → Kể tự do
RESULT	**Notes saved after the user confirms**
RULES	BR-01 · consent, Gemini JSON, edit, fallback
SOLVES	Slow evenings

WF-02

Publish a user card to the shared catalog

ACTOR	User + Staff
TRIGGER	User shares an own card
RESULT	**Card in the catalog, app updates, inbox message**
RULES	BR-02 submit · BR-03 approve / reject + reason
SOLVES	Static content

WF-03

Announce news to every user

ACTOR	Staff
TRIGGER	Staff has news for everyone
RESULT	**Announcement published; every inbox shows it**
RULES	BR-04 title 1–80, body 1–1000, staff only
SOLVES	No channel

Supporting features: onboarding & starter plan · weekly card PDF · streaks & journey · self-control & breathing · groups · Discord link · dashboard · roles.

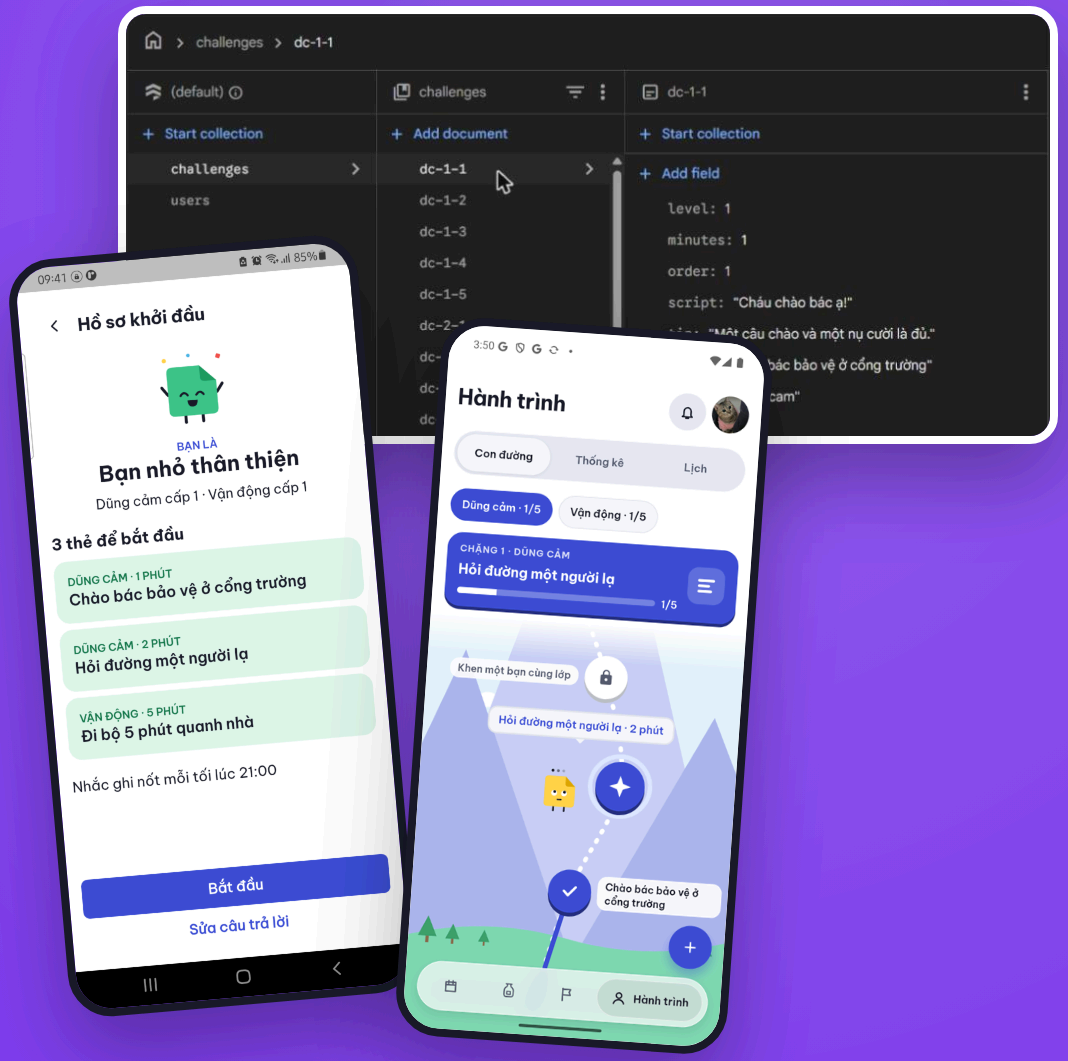
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02

Design

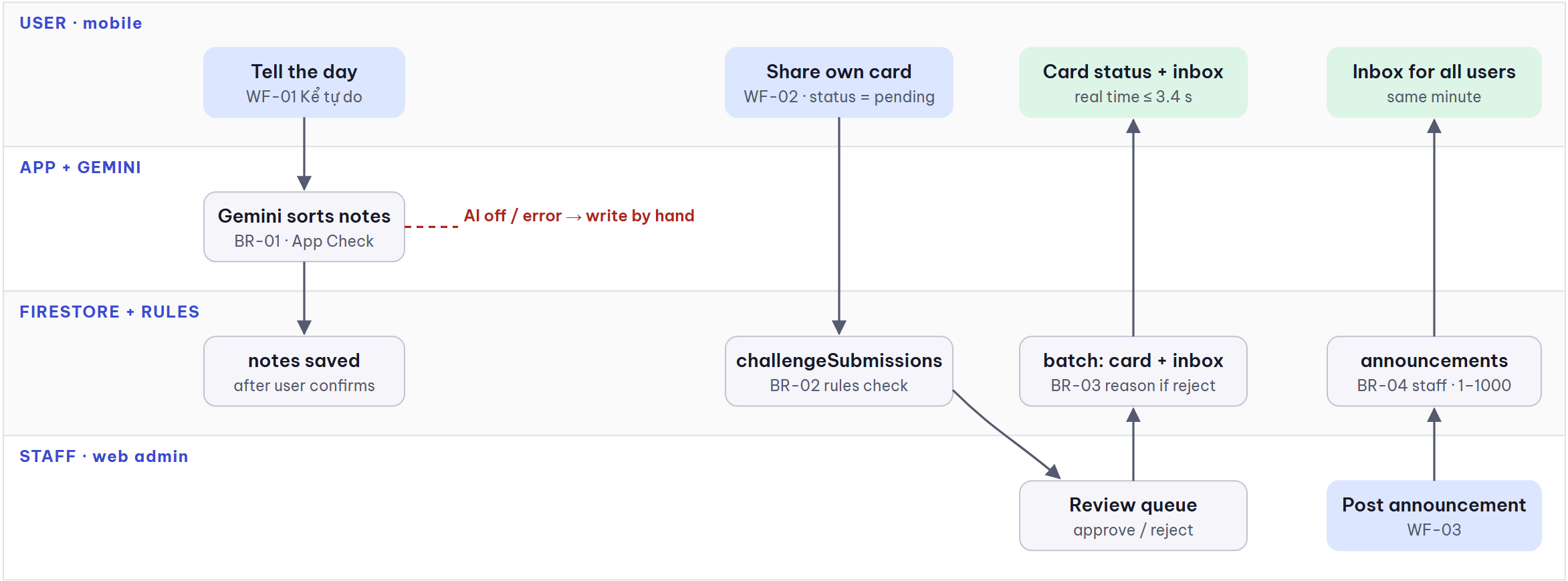
How is the whole system built?

07 Business flow | 08 State machine | 09 Architecture | 10 Stack | 11 UI/UX | 12 Sequence | 13 Classes | 14–15 Database | 16 Traceability



Business flow in one swimlane view

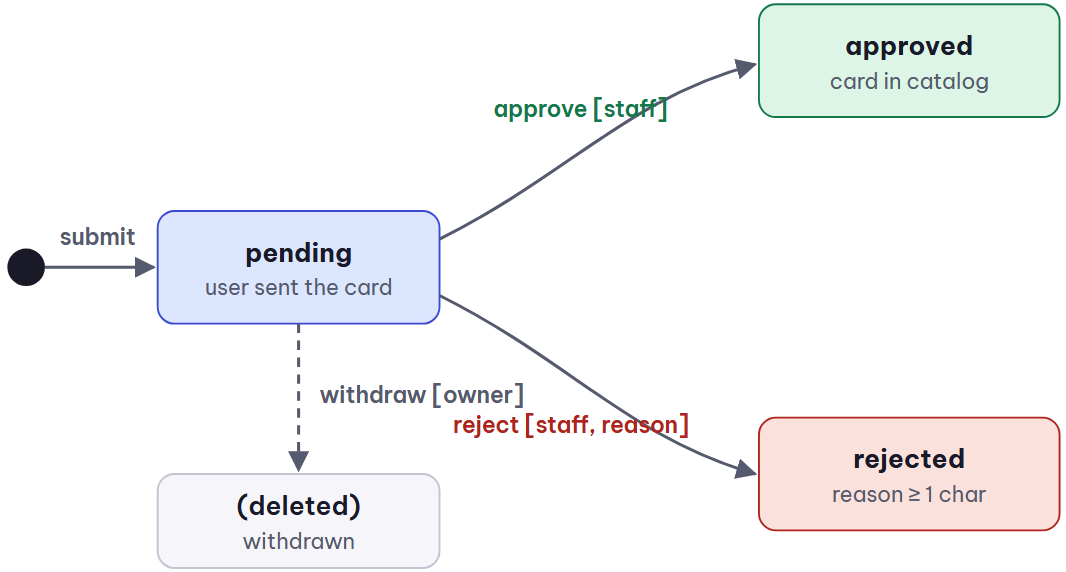
The three workflows share one Firestore; rules are the decision points.



BR-01 AI only when enabled, consented and App Check passes · BR-02 user creates pending only · BR-03 reject needs a reason · BR-04 announcements: staff only, title 1–80, body 1–1000 · BR-05 only admins change roles, never their own.

A shared card: pending → approved or rejected

Entity: challengeSubmissions/{id}. Every transition and every forbidden one has a rules test.



Forbidden (rules deny)

user → approved
approved → pending
reject without reason
withdraw after review

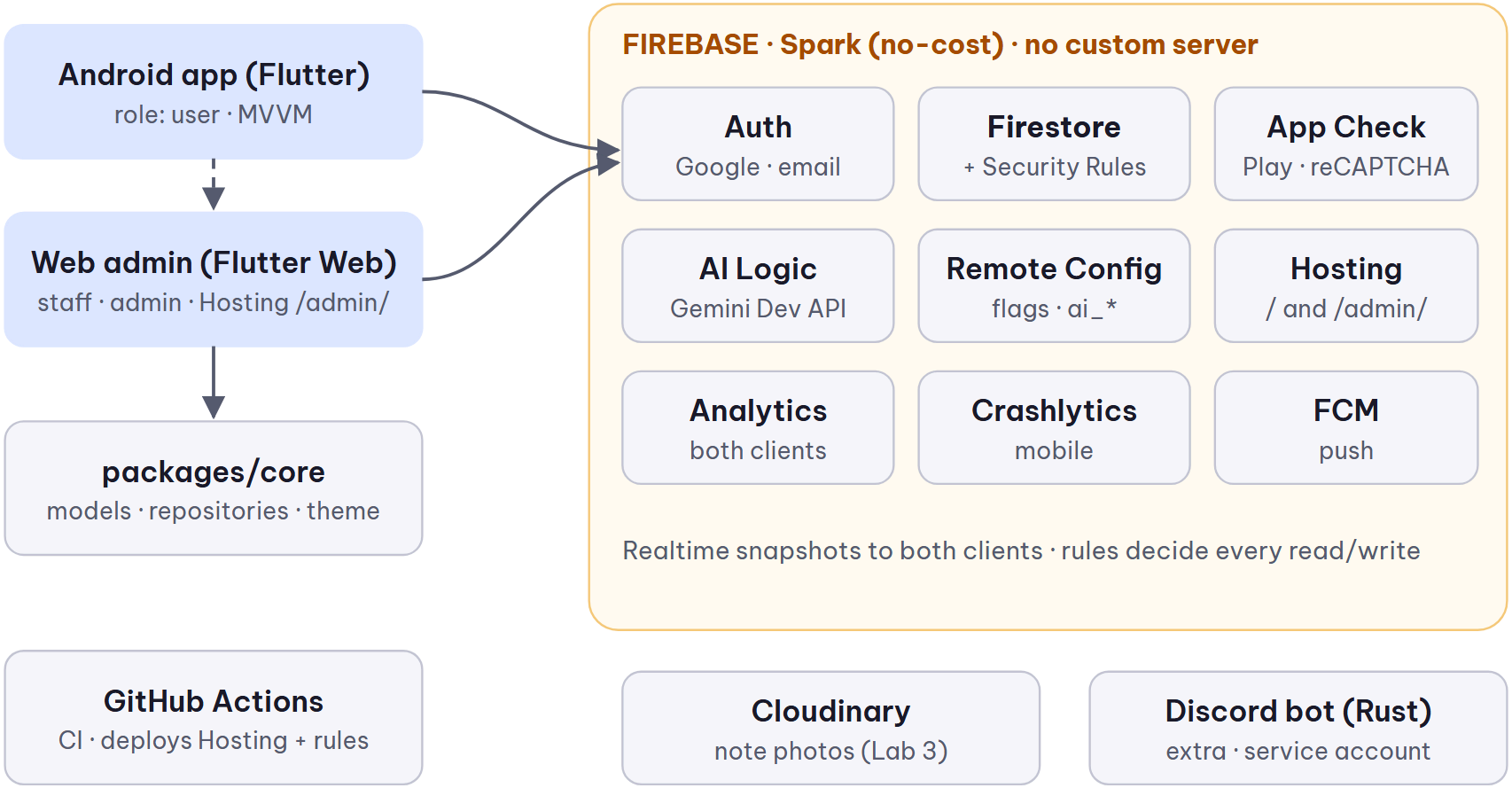
```
109                     && request.resource.data.createdAt == request.time;
110 ...         // Only pending -> approved | rejected; a rejection needs a reason.
111             allow update: if isStaff()
112                     && resource.data.status == 'pending'
113                     && request.resource.data.status in ['approved', 'rejected']
114                     && request.resource.data.diff(resource.data).affectedKeys()
115                         .hasOnly(['status', 'reason', 'reviewedAt', 'reviewedBy'
116                     && request.resource.data.reviewedBy == request.auth.uid
117                     && request.resource.data.reviewedAt == request.time
118                     && shortText(request.resource.data.get('reason', ''), 0, 200)
119                     && (request.resource.data.status == 'approved'
```

firestore.rules · the guard on every review (BR-03)

Tests: R-20 create pending only · R-21 read own or staff · R-22 staff review, reason on reject · R-23 withdraw while pending. 30/30 rules tests pass.

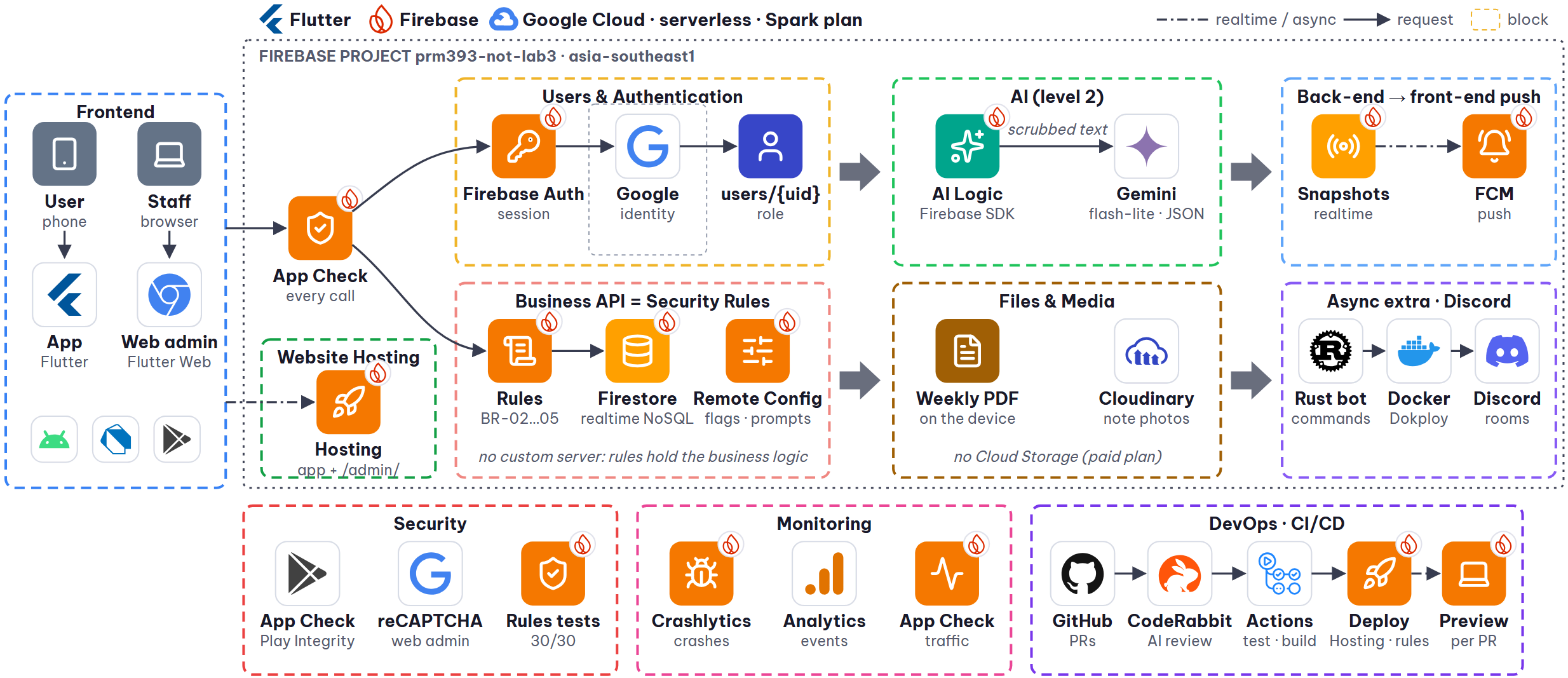
Two Flutter clients, one Firebase backend

Decision: no custom server. Security Rules + App Check are the backend's access layer.



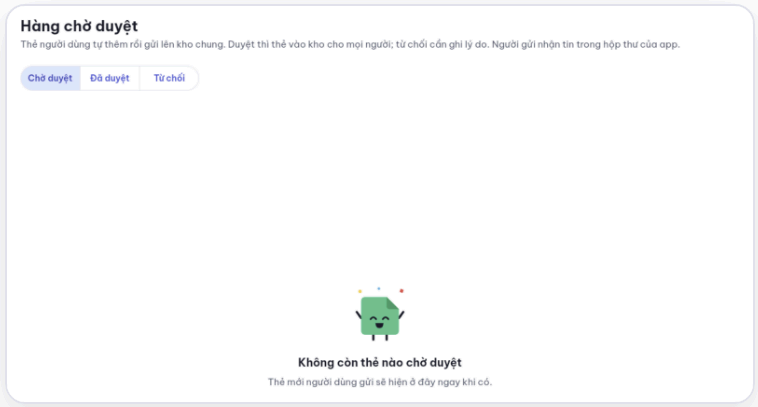
Deploy: GitHub Actions builds both clients, deploys Hosting (/ and /admin/) and Firestore rules + indexes on merge to main.

Serverless on Firebase: Security Rules are the backend

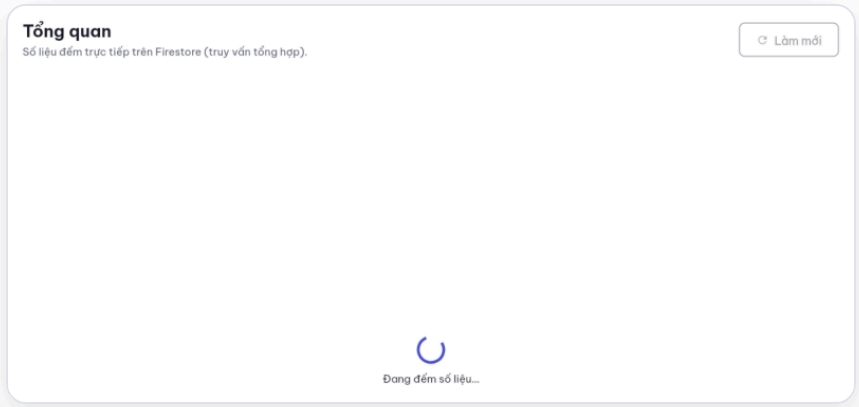


One design system on phone and web

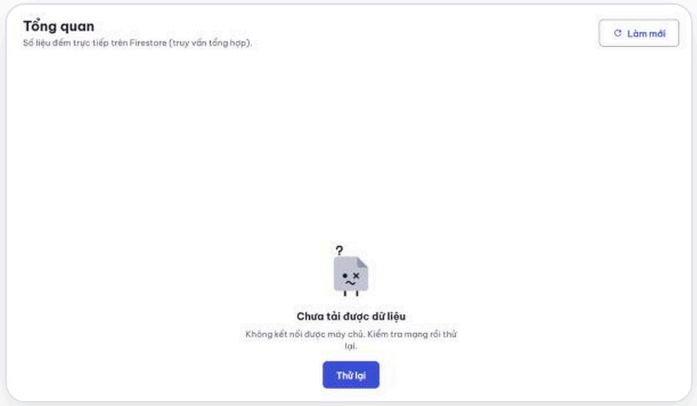
Lab 2 tokens live in packages/core; the web admin reuses them in a wide layout.



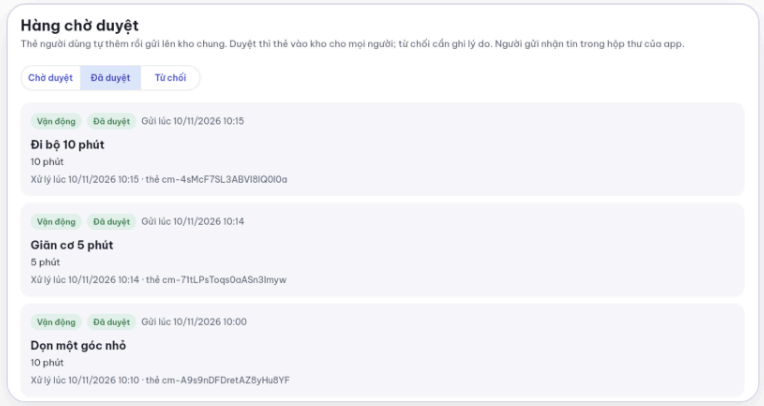
Empty · Hàng chờ duyệt



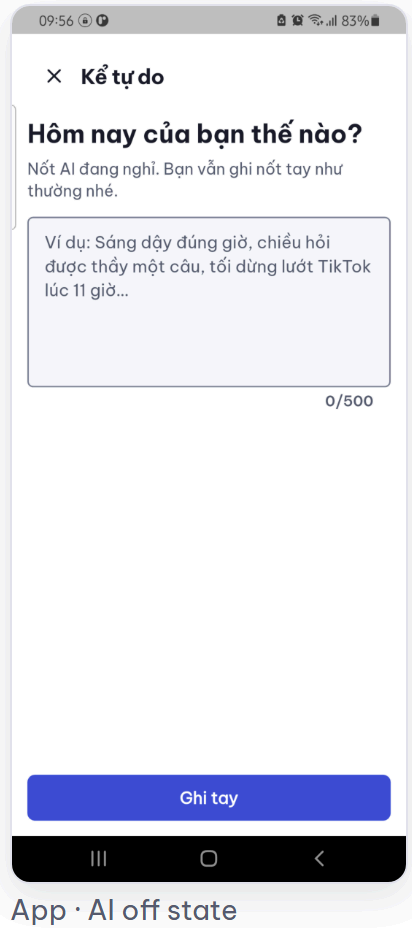
Loading · Tổng quan



Error · server unreachable, Thử lại



Success · approved cards



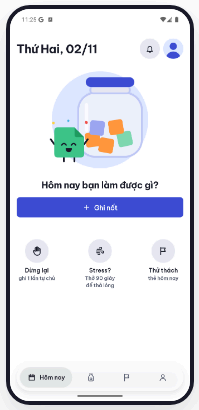
App · AI off state

States: empty · loading · error (within 15 s, Thử lại) · success · wrong role rejected on web and mobile.

Screen map: twelve screens, two clients

Real screenshots from the Galaxy A70 and the deployed web admin; codes match docs/design-traceability.md.

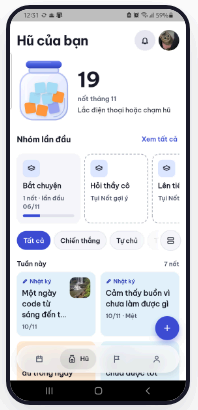
ANDROID · USER



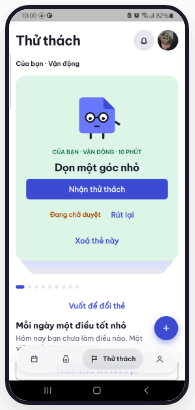
S02 Today



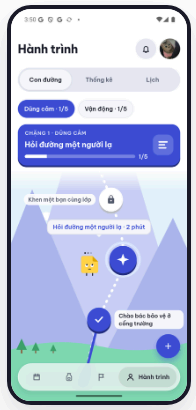
S03b Nốt AI



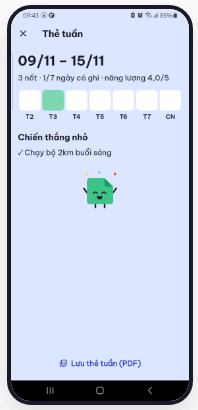
S05 Jar



S07 Challenges



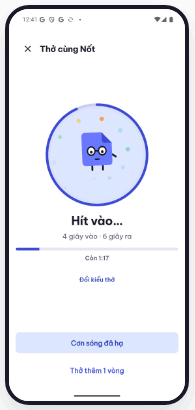
S11 Journey



S11b Weekly PDF

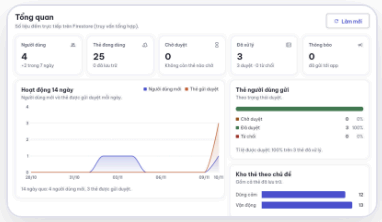


S13 Onboarding

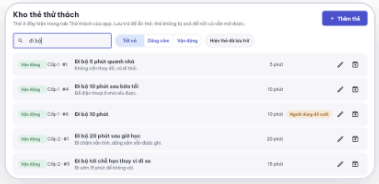


S10 Breathing

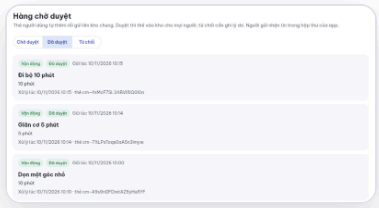
WEB · STAFF



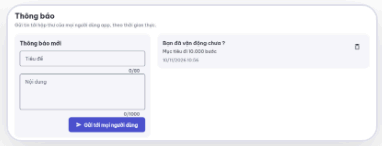
W1 Dashboard



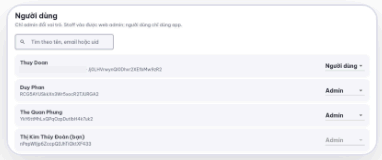
W2 Catalog



W3 Review queue



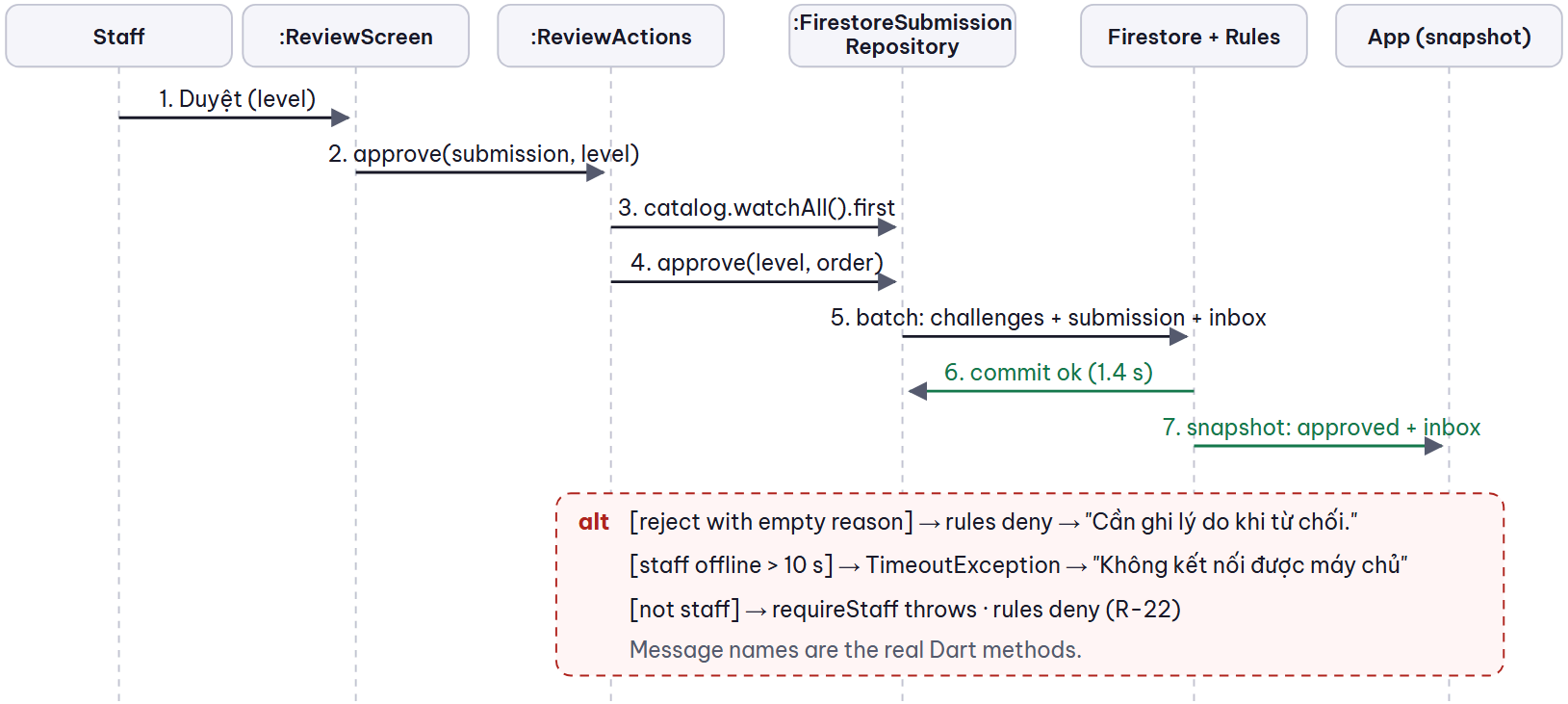
W4 Announcements



W5 Users & roles

Approving a card is one batch write

Security Rules guard the write; snapshots carry the change back to the phone.




```
156          })
157          ..update(_col.doc(submission.id), <String, Object?>{
158            'status': SubmissionStatus.approved.id,
159            'reason': '',
160            'challengeId': cardId,
161            'reviewedBy': reviewerUid,
162            'reviewedAt': FieldValue.serverTimestamp(),
163          })
164          ..set(_inbox(submission.uid).doc(), <String, Object?>{
165            'kind': InboxKind.submissionApproved.id,
166            'title': 'Thẻ của bạn đã vào kho chung',
167            'body': _clip(
168              '"${submission.title}" đã được duyệt. Mọi người giờ có thể nhận '
169              'thẻ này ở tab Thử thách.',
170              500,
171            ),
172            'submissionId': submission.id,
173            'read': false,
174            'createdAt': FieldValue.serverTimestamp(),
175          });
176        await batch.commit();
177      });
```

FirestoreSubmissionRepository.approve: one batch

MVVM: screens never touch Firebase

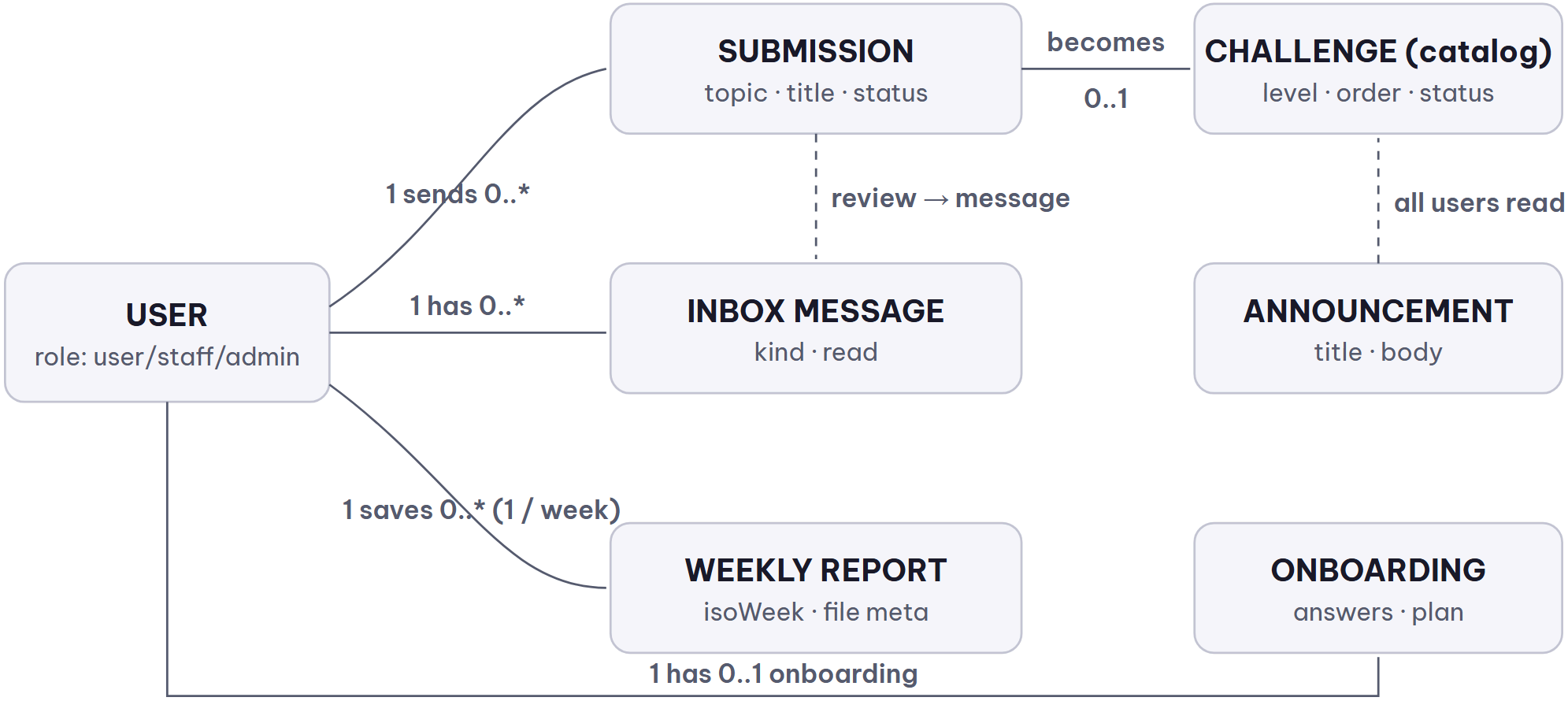
Business rules sit in the repository + Security Rules, not in widgets.

LAYER	WF-01 NỐT AI	WF-02 SHARE & REVIEW	WF-03 ANNOUNCE
Screen	`FreeTellScreen`	`ChallengesScreen` · `ReviewScreen`	`AnnouncementsScreen` · `NotificationInboxSheet`
ViewModel	`FreeTellViewModel`	`CommunityActions` · `ReviewActions`	`AnnouncementActions` · `communityInboxProvider`
Service	`AiService`	`SubmissionRepository` · `CatalogRepository`	`AnnouncementRepository` · `InboxRepository`
Implementation	`FirebaseAiService`	`FirestoreSubmissionRepository`	`FirestoreAnnouncementRepository`
Model	`AiSortResult` · `AiNoteSuggestion`	`ChallengeSubmission` · `CatalogCard`	`Announcement` · `InboxMessage`

main.dart wires each provider to its Firebase implementation; tests swap in fakes (303 app · 14 web admin · 5 core tests).

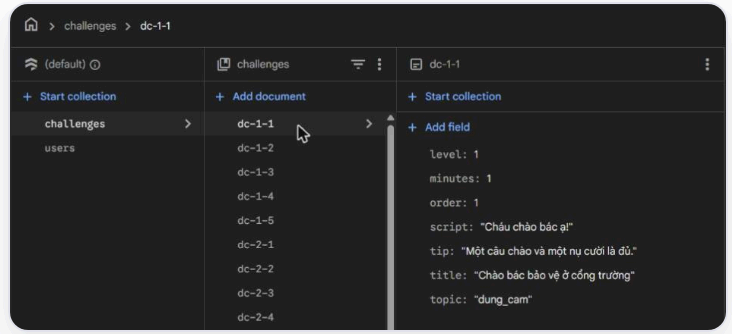
Six new entities around the user

Lab 3 entities (notes, habits, groups, progress) stay private to their owner.

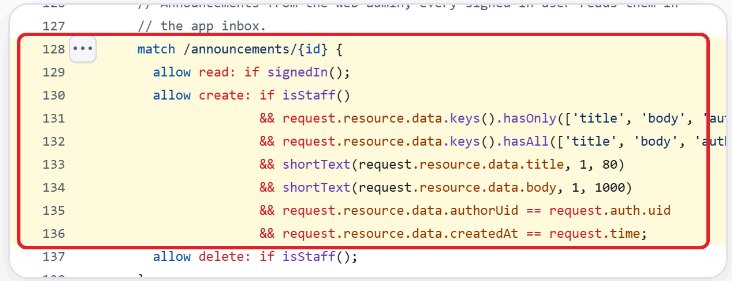


Firestore collections and their rules

COLLECTION	KEY FIELDS	CONSTRAINTS (RULES)
users/{uid}	role, displayName, email	self-create as user; only admin changes role, never own
challenges/{id}	topic, level 1–10, order, title ≤ 80, status	staff only; updatedBy = auth.uid; archive, no delete
challengeSubmissions/{id}	uid, topic, title, minutes, status, reason	create pending; staff review; reason on reject
users/{uid}/inbox/{id}	kind, title, body, read	staff create; owner reads, may only flip read
announcements/{id}	title ≤ 80, body ≤ 1000, authorUid	staff create/delete; every signed-in user reads
users/{uid}/reports/{week}	isoWeek, fileName, sizeBytes, noteCount	owner only; id = week; savedAt = server time
users/{uid}/private/onboarding	answers, profileLabel, planSource	owner only; enums and sizes checked



Firestore Console · challenges



Rules for announcements (BR-04)

Indexes: submissions (uid, createdAt↓), (status, createdAt↓) · notes (groupId, createdAt↓) · full table: docs/data-model.md

Traceability: screen → rule → test

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WORKFLOW	SCREENS	CLASS / SERVICE	COLLECTION	TESTS
WF-01 Nốt AI	S03b Kể tự do	FreeTellViewModel · AiService	notes	BR-01 · free_tell_flow_test · ARTEMIS T18 (2/3), T19 (3/3)
WF-02 Share & review	S07 · Review queue · Inbox	CommunityActions · ReviewActions	challengeSubmissions · challenges · inbox	BR-02/03 · R-20…R-24 · ARTEMIS T20 (3/3) · D-1…D-3
WF-03 Announce news	Announcements · Inbox	AnnouncementActions	announcements · inbox	BR-04 · R-25 · notification_inbox_test · web checklist #9–11
Weekly PDF	S11b Thẻ tuần	WeeklyReportViewModel	reports	R-28 · weekly_report_test · ARTEMIS T14 (3/3)
Onboarding	S13a–g	OnboardingViewModel	private/onboarding	R-27 · onboarding_flow_test · ARTEMIS T17 (3/3)
Roles	Sign-in · Users	sessionProvider · UserActions	users	BR-05 · R-01, R-26, R-29 · session_test · auth_flow_test

Full matrix: docs/design-traceability.md · rules: firebase/tests/firestore.rules.test.js · E2E: e2e_test/

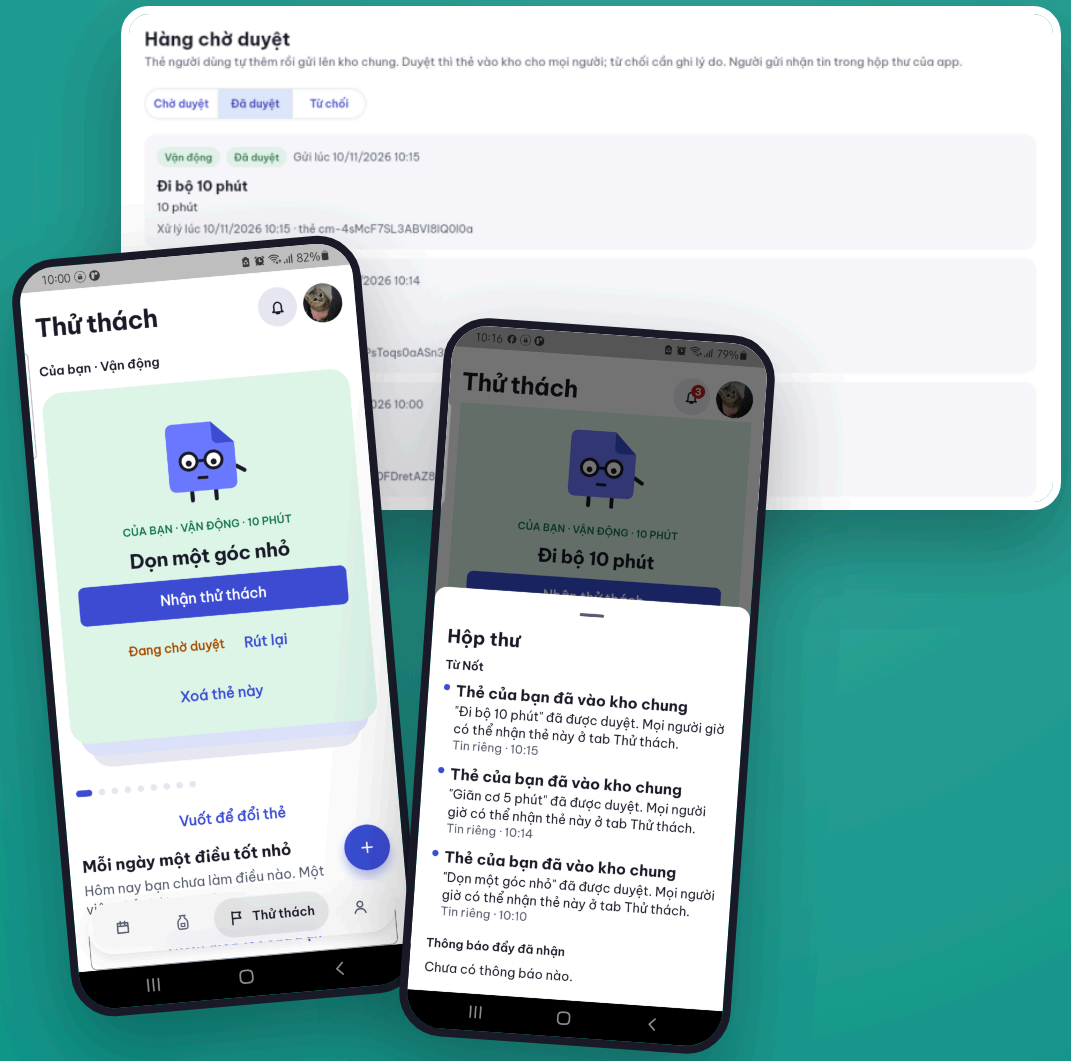
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03

Demo

How do the workflows run for real?

17 Demo plan | 18 WF-01 Nốt AI | 19 WF-02 Publish a card | 20 WF-03 Announce



Demo plan: one evening

Minh logs his day and shares a card; staff publish it and announce. Each result feeds the next workflow.

01 · USER
Kể tự do: one paragraph → 3 notes saved

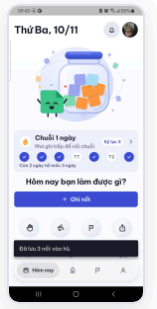
→

02 · USER → STAFF
Share "Đi bộ 10 phút" → staff approves → app shows it

→

03 · STAFF → USERS
"Bạn đã vận động chưa?" → every inbox

RESULT WF-01



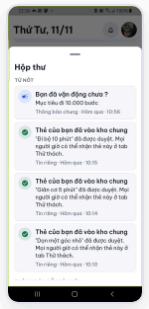
Jar 11 → 14 notes, streak starts

RESULT WF-02



Card "Người dùng đề xuất" in catalog; inbox message

RESULT WF-03



Announcement in app inbox at 10:56, same minute

EXCEPTION PATHS SHOWN LIVE

- Remote Config `ai_enabled = false` → "Nốt AI đang nghỉ", user writes by hand
- Reject without reason → blocked (BR-03) · user account on the web, staff account on the phone → rejected (BR-05)

Prepared: user account on a Galaxy A70 (release build, Play Integrity) · staff/admin account on the web admin · ai_enabled ready to toggle in Remote Config · empty review queue · backup video.

WF-01 Log the day with Nốt AI

One paragraph becomes three editable notes.

TRIGGER · PRECONDITION

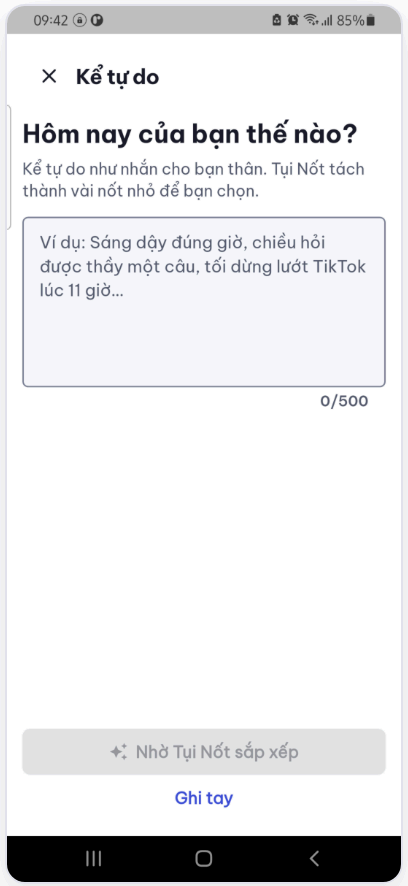
Evening, Ghi nốt → Kể tự do · role `user`, AI on (BR-01)

STEPS → RESULT

- Consent sheet on first use
- Scrubbed paragraph → Gemini → JSON notes + mood
- Labelled "Gợi ý từ Nốt AI", editable
- **Result:** chosen notes saved, jar and streak update

EXCEPTION

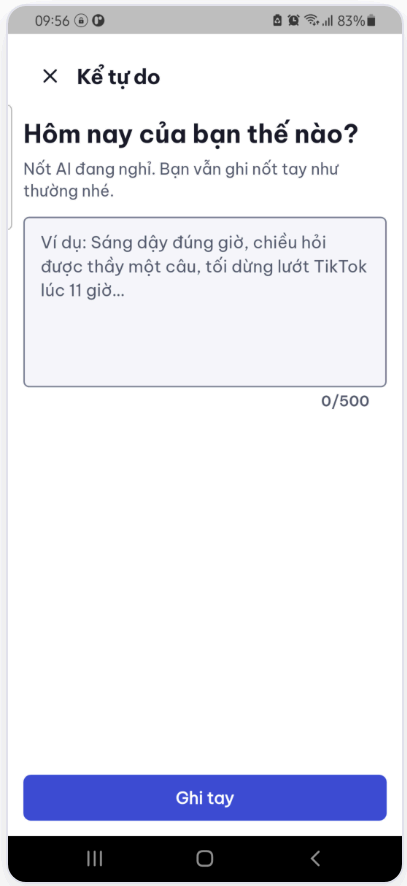
BR-01 fails (AI off, App Check, network) → "Ghi tay", paragraph kept



Consent



Suggestions

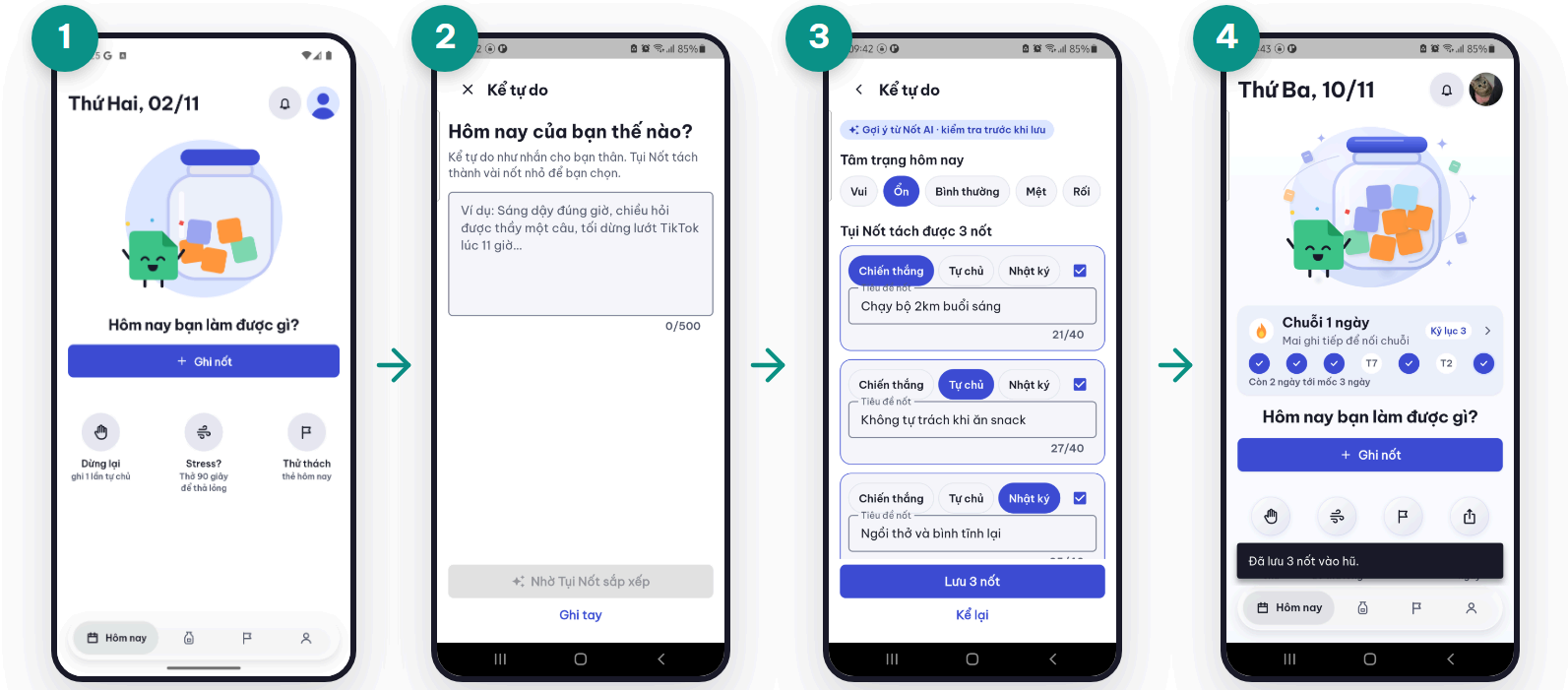


ai_enabled = false

Evidence: Galaxy A70, real Gemini (~2 s) · ARTEMIS T18 2/3, T19 (AI off) 3/3 · AI level 2: JSON schema → AiSortResult.

WF-01 in four screens

From an empty evening to three saved notes.



Ghi nốt
evening, from Today

Consent
what is sent to Gemini

Suggestions
3 notes + mood, editable

Saved
jar 11 → 14, streak starts

Exception shown live: ai_enabled = false → frame 2 says 'Nốt AI đang nghỉ' and only 'Ghi tay' is left.

WF-02 Publish a card

Staff approve on the web; the phone updates within seconds.

TRIGGER · STATE

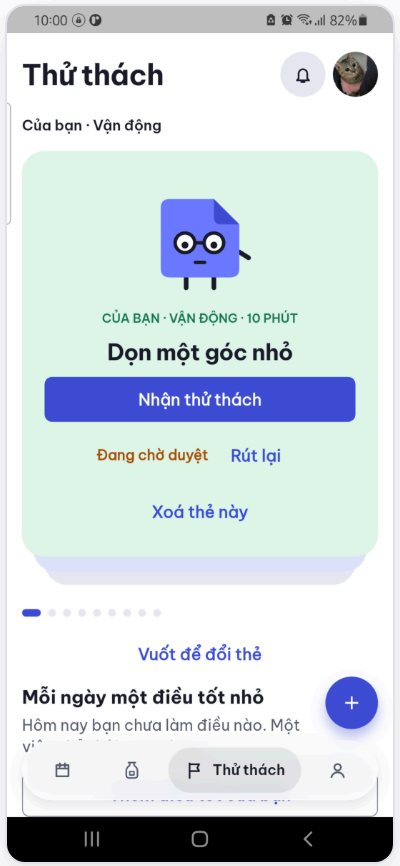
User shares own card · **pending** → **approved** / rejected

STEPS

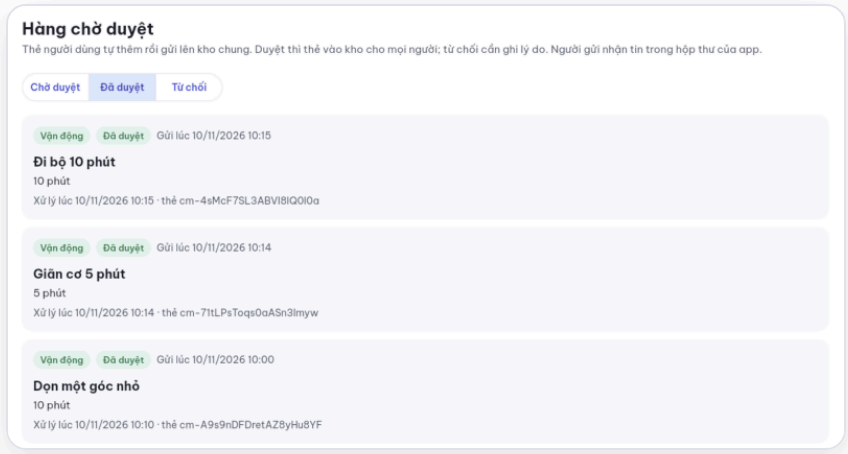
- User shares → pending
- Staff: Duyệt on the web
- Batch: card + status + inbox
- App: "Đã vào kho chung ✓"

EXCEPTION

BR-03 reason required · BR-02 user cannot approve



Đang chờ duyệt



Web: approved

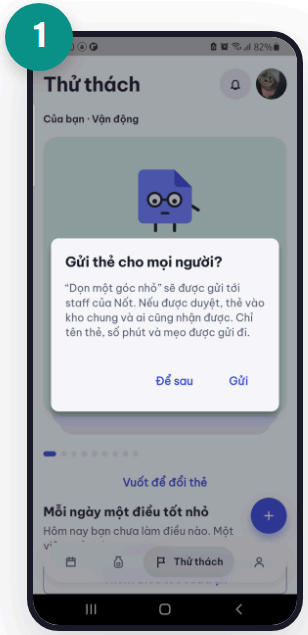


App: inbox

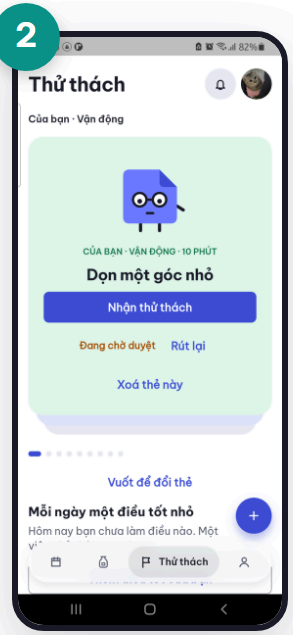
Evidence: write 1.4 s after Duyệt · phone ≤ 3.4 s (D-2, D-3), 22 s when idle (D-1) · ARTEMIS T20 3/3 · docs/cross-role-flow.md

WF-02 in five screens

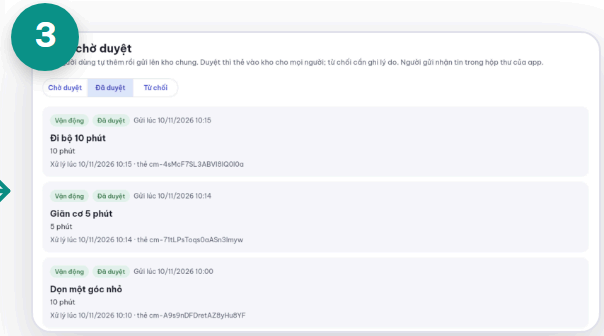
A private card becomes everyone's card.



Share
user sends own card



Pending
"Đang chờ duyệt"



Approve
staff, web admin



In catalog
"Đã vào kho chung ✓"

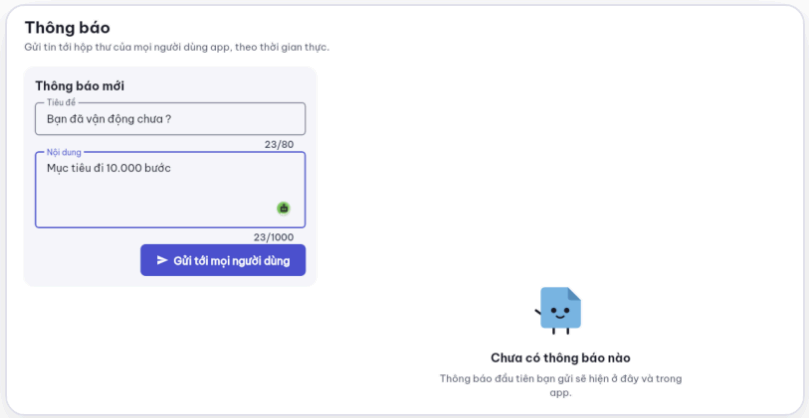


Inbox
message for the author

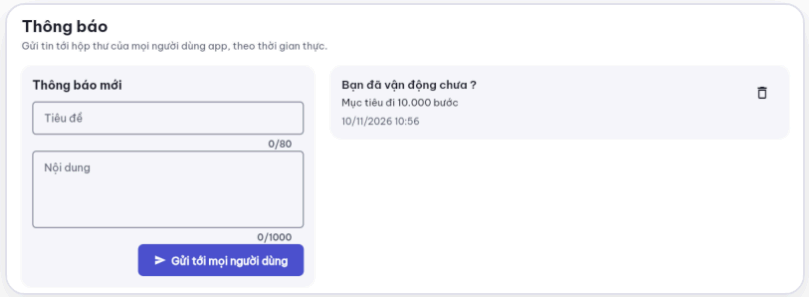
Exception shown live: Từ chối without a reason is blocked (BR-03); the card stays pending.

WF-03 Announce news to every user

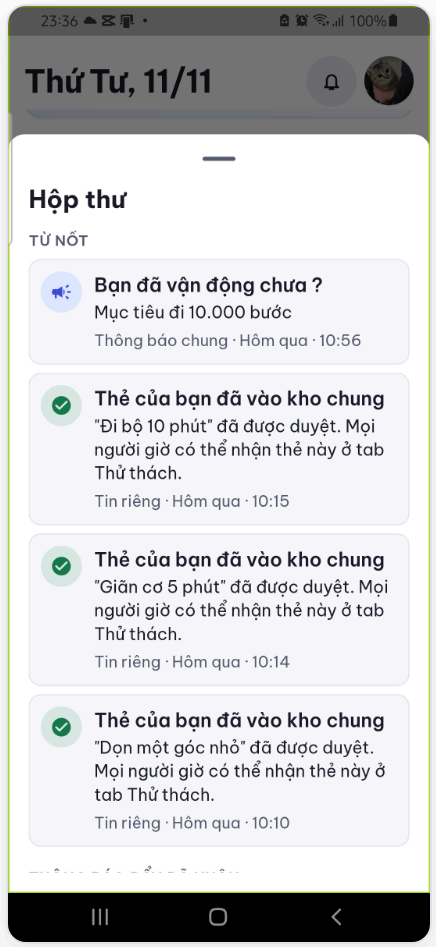
Staff post once; every inbox shows it in the same minute.



1 · Compose (list still empty)



2 · Sent at 10:56, form cleared



3 · App inbox, 10:56

TRIGGER · PRECONDITION

Staff has news for every user · signed in as staff or admin

DECISION (BR-04)

Title 1–80 · body 1–1000 · authorUid = self, in the form and in the rules

RESULT

Announcement published; every user's inbox shows it in real time

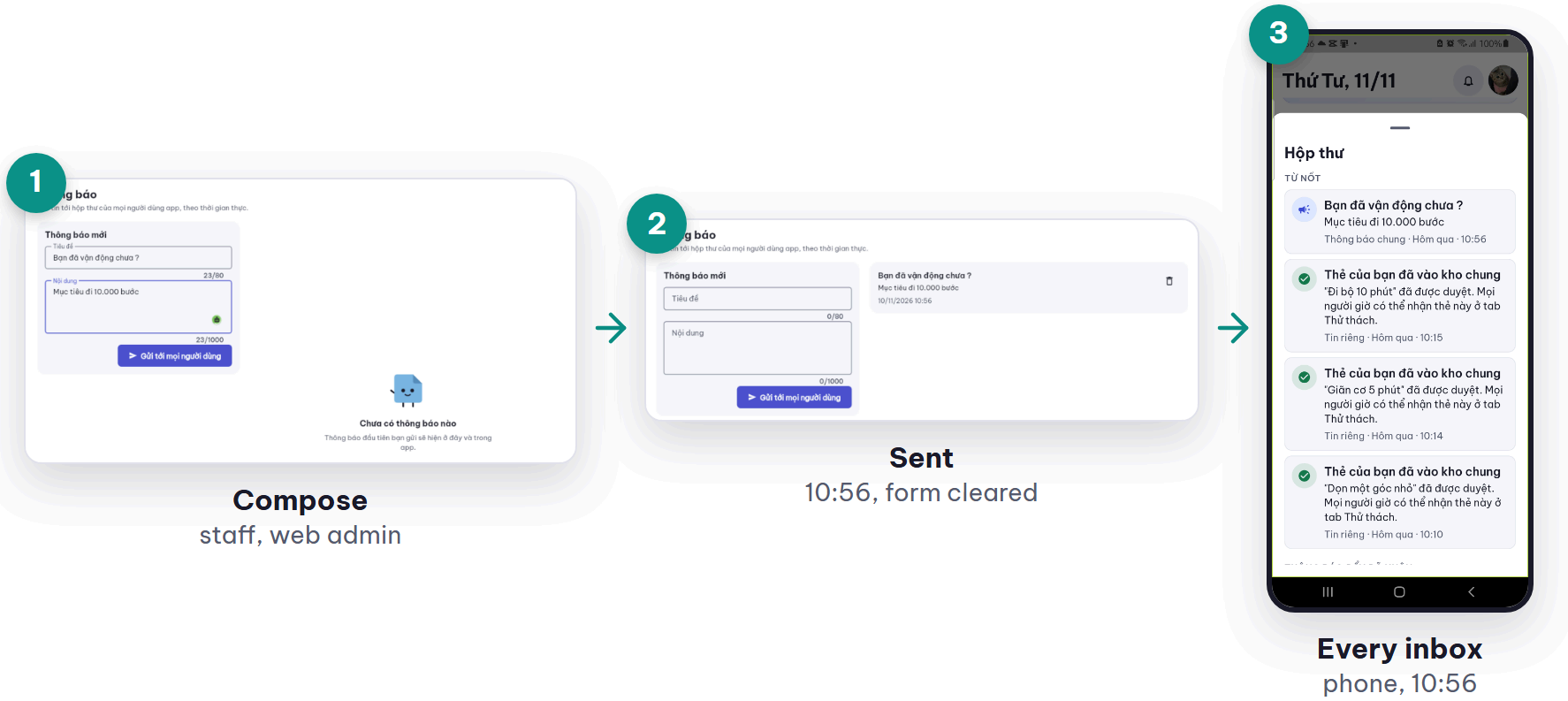
EXCEPTION

Empty or too long → form error, nothing saved · user account → rejected at sign-in, rules deny (R-25)

Evidence: sent at 10:56, in the app inbox at 10:56 · rules R-25 (staff only, authorUid = self, title 1–80, body 1–1000) · web checklist #9–11.

WF-03 in three screens

One form, every inbox.



Compose
staff, web admin

Sent
10:56, form cleared

Every inbox
phone, 10:56

Exception shown live: empty body or > 1000 characters → form error, nothing saved (BR-04).

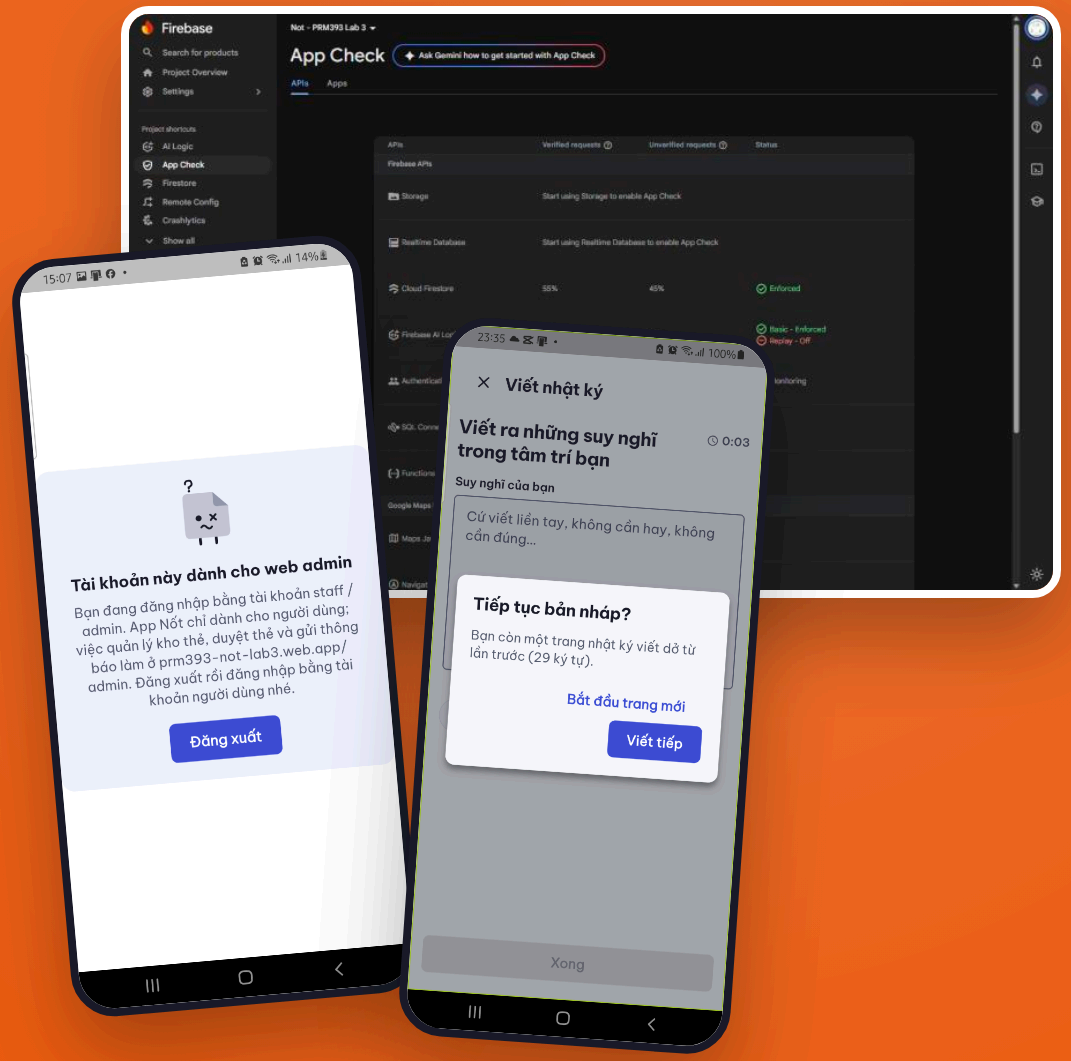
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04

Proof

What can we prove?

21 Testing | 22 Achievements | 23 Limitations | 24 Conclusion



Testing at three levels

Rules, code and the deployed system are each tested, with evidence in the repo.

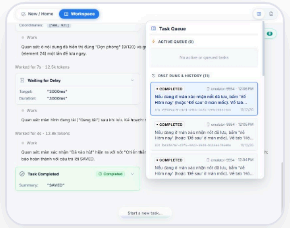
30/30
Security Rules tests

303
Flutter app tests

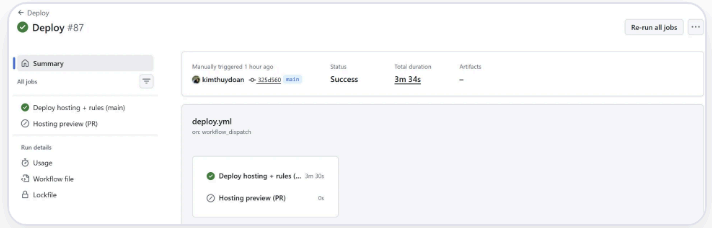
14 + 5
web admin + core tests

14/15
ARTEMIS Final runs, real phone

LEVEL	EVIDENCE	RESULT
Rules	Emulator, allowed + denied per role per collection (BR-02…BR-05)	30 / 30
Unit / widget	ViewModels, repositories, screens at 360 dp	all pass
Cross-role	3 runs on the deployed system, timed	3 / 3
Web checklist	Actions, rejection, empty, loading, error, live role change	16 / 16
Real device	Onboarding, Nốt AI, PDF, AI off, journal draft, role rejection on Galaxy A70	pass
E2E (ARTEMIS)	Lab 3: 41/42 · Final T14, T17–T20 on Galaxy A70: 14/15 · T15/T16 not run	14 / 15



ARTEMIS: agent reports SAVED

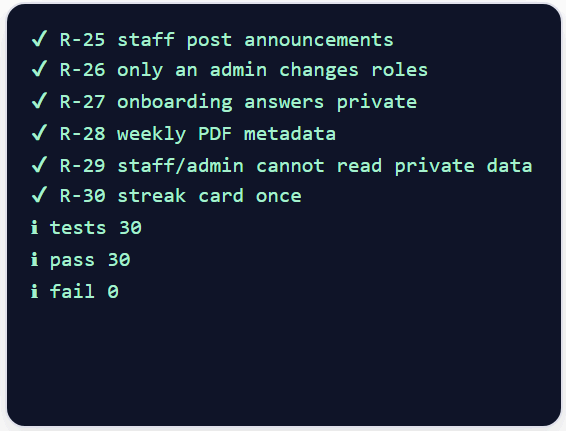


GitHub Actions: deploy on main

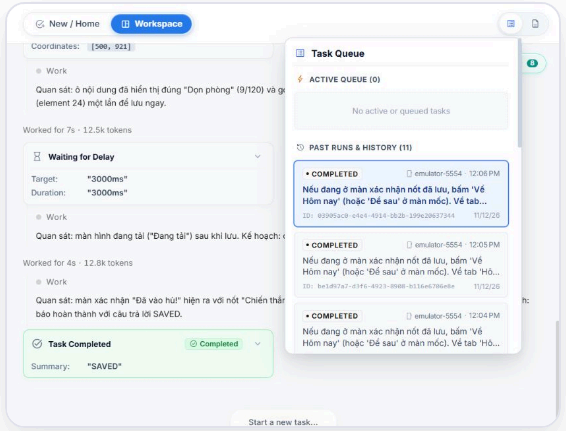
CI on every PR: format, analyze, all test suites, rules in the Emulator, web build, Android build, Hosting preview · details: docs/test-report.md.

Evidence wall

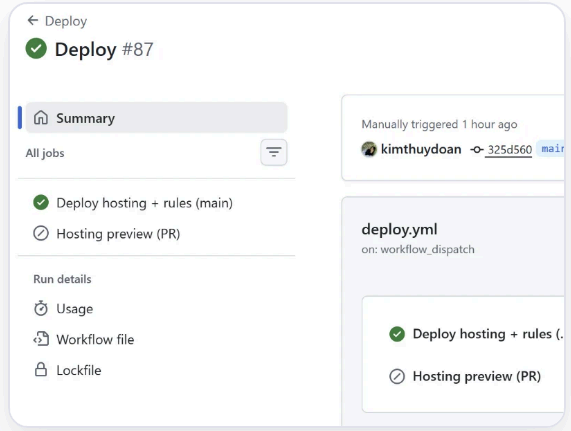
Every number on the testing slide has a screenshot or a log behind it.



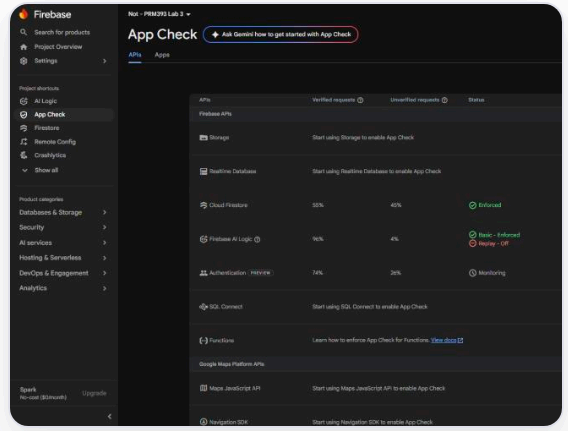
Security Rules · Emulator 30/30



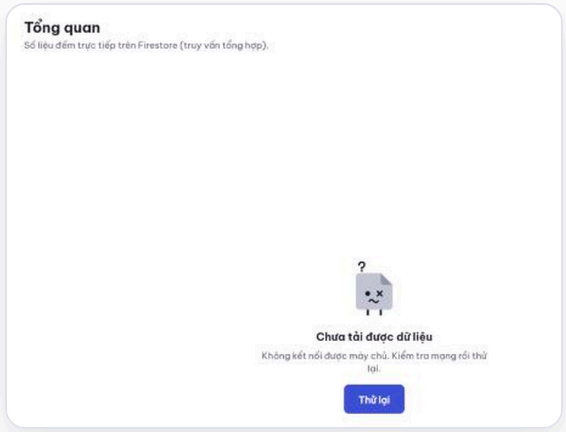
ARTEMIS on a real phone 14/15



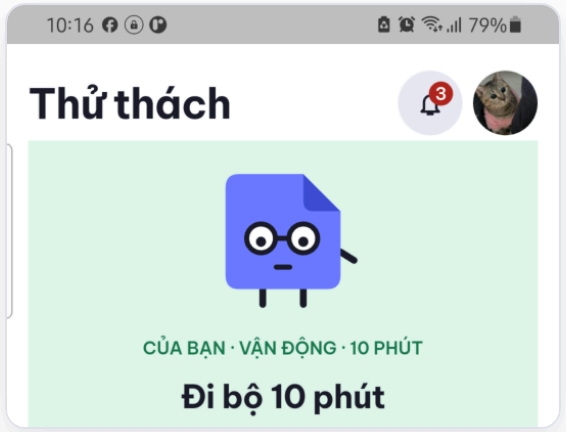
GitHub Actions deploy green



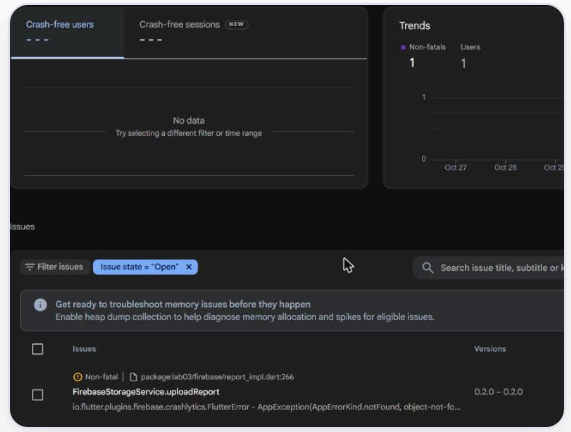
App Check enforced



Web checklist · error state 16/16



Cross-role, deployed 3/3



Crashlytics live



Analytics events live

All files: evidence/ in the repo · details: docs/test-report.md

P4 22 ACHIEVEMENTS

Achievements

Every brief requirement maps to a live feature and a piece of evidence.

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BRIEF REQUIREMENT	IMPLEMENTED	EVIDENCE
Web admin: 4 screens + announcement form, deployed	Dashboard, catalog, review, announcements, users	`evidence/web-admin/` (16 items)
Role-based rules, deny by default, tested	user / staff / admin, R-01…R-30	`evidence/firestore/rules-tests.txt`
Cross-role flow, real time, documented	Share card → approve → app + inbox	`docs/cross-role-flow.md`, `evidence/cross-role/`
AI level 2 with RC, App Check, fallback	Kể tự do on Gemini, `ai_*` flags incl. prompts, enforced App Check	`docs/ai-feature.md`, ARTEMIS T18/T19
App Check on Firestore and AI Logic	**Enforced** on both · Web: reCAPTCHA Enterprise; Android: Play Integrity	`evidence/app-check/`
Lab 3 PDF saved & shared on device	pdf + share_plus, metadata in Firestore	15.4 KB file pulled from the phone
Mobile data on Firestore with inbox	Announcements + personal inbox, real time	web checklist #11

Not met: ARTEMIS T15/T16 (email sign-in) not run → slide 23

Limitations and next steps

What is not done yet, and what it costs.

LIMITATION	IMPACT	PRIORITY	NEXT STEP
ARTEMIS T15/T16 (email sign-in) not run	Sign-out / sign-in by email has widget tests only, no E2E on a device	Medium	Create a test-only account, run 3× each
Names can reach Gemini (CR-F2)	Email and phone are scrubbed, but a friend's name is sent as typed	Medium	Consent sheet says so; add on-device name masking
Exploratory session X01: 10 UI issues not rechecked	Possible small layout bugs on rarely used screens	Low	Re-run X01 on the Final build, fix or close each
D-1 took 22 s on an idle phone	Real time not guaranteed after long idle	Low	Measure more runs; re-listen on resume
Lab 2 flow A3 (practice chat) dropped	Overlaps the Discord rooms; one designed flow not built	Low	Reason in docs/gap-review.md §1.1

Conclusion

Nốt is now a working multi-role system, and we can show every claim.

1 · Problem solved

Content is curated by staff and grows from users' own cards; news reaches every user in the same minute.

2 · Working product

WF-01, WF-02, WF-03 run live on Android and the deployed web admin, on the free plan.

3 · Verified

30/30 rules tests, 300+ code tests in CI, 14/15 ARTEMIS runs on a real phone, 3 timed cross-role runs, App Check enforced.

prm393-not-lab3.web.app

prm393-not-lab3.web.app/admin/

github.com/FU-PRM-Projects/Lab-03

Thank you · Questions?

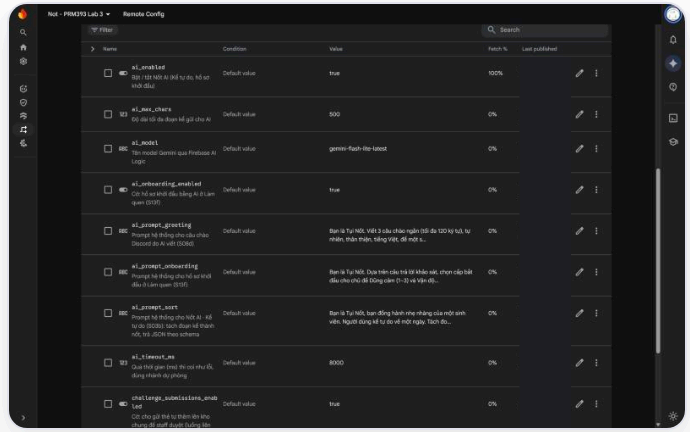
AI examples

Three real Gemini inputs and outputs, and one handled failure.

INPUT (KỂ TỰ DO)	GEMINI OUTPUT
Ran 2 km, ate a snack I meant to skip, calmed down without blaming myself	chien_thang "Chạy bộ 2km quanh hồ" · nhat_ky "Lỡ ăn thêm gói snack..." · tu_chu "Không tự trách và bình tĩnh lại" · mood binh_thuong
Asked a stranger in the library where to charge my phone; wrote a few lines about the rainy afternoon	chien_thang "Lấy hết can đảm hỏi mượn chỗ sạc pin" · nhat_ky "Ghi lại vài dòng cảm xúc buổi chiều mưa" · mood vui
Tired for days, only tidied my desk today	chien_thang "Mình đã dọn dẹp được cái bàn học" · 2× nhat_ky · mood met · needs_support false
Handled failure	404 model retired / 401 missing App Check → `ai_error` logged, paragraph kept, "Ghi tay" offered; fixed by Remote Config model name

```
29
30  GenerativeModel _model(String instructions, Schema schema) =>
31      FirebaseAI.googleAI().generativeModel(
32        model: config.model,
33        systemInstruction: Content.system(instructions),
34        generationConfig: GenerationConfig(
35          responseMimeType: 'application/json',
36          responseSchema: schema,
37          temperature: 0.4,
38        ),
39      );
```

JSON schema forces the answer shape

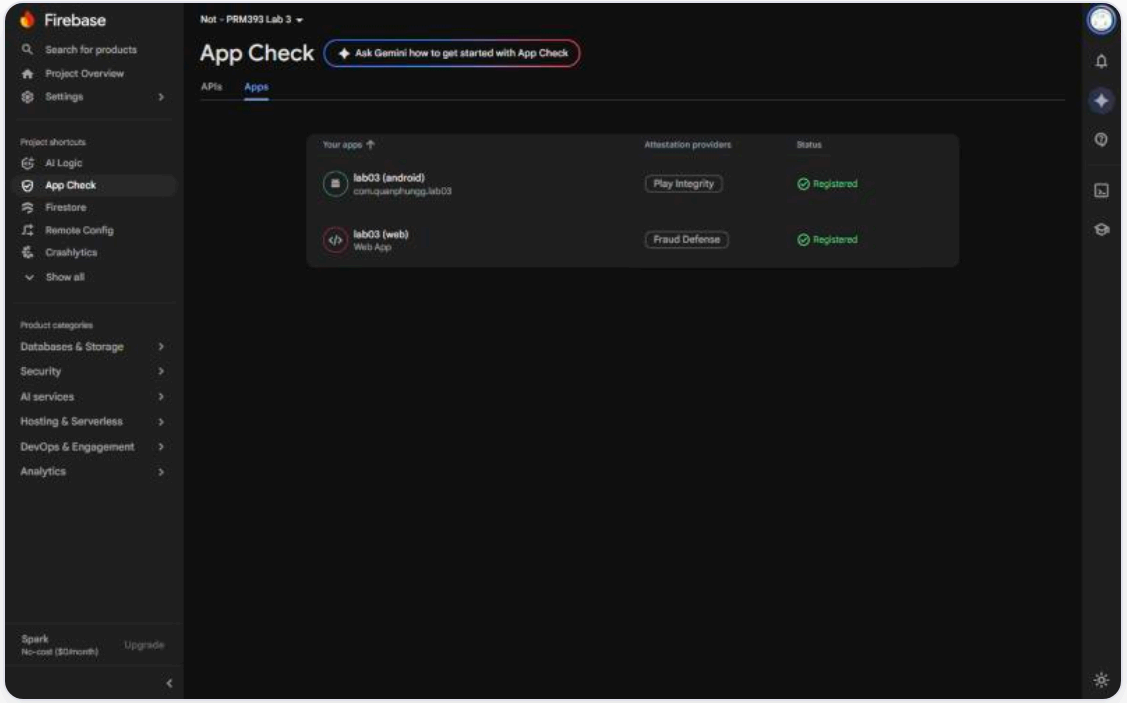


Remote Config: model, prompts, on/off

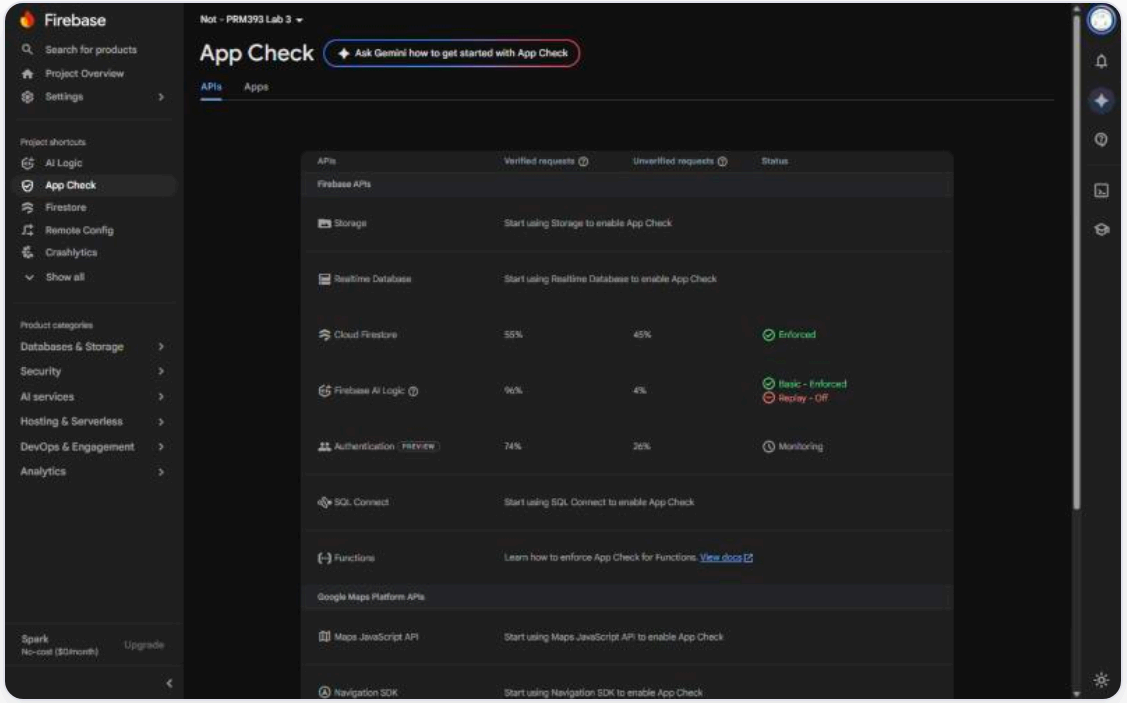
Inputs are scrubbed (email, phone) and capped at 500 characters; no names, emails or photos are sent.

App Check: registered and enforced

Both apps are registered; Firestore and AI Logic are enforced.



Apps: lab03 (android) Play Integrity · lab03 (web) Fraud Defense



Cloud Firestore: Enforced · AI Logic: Enforced

Web: reCAPTCHA Enterprise (Fraud Defense) · Android: Play Integrity (debug builds need a registered debug token) · enforced 12/11/2026 · evidence/app-check/.

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Lessons learned

Running the system for real found five bugs the tests did not; all are fixed, with regression tests.

APPROVE TIMED OUT

Riverpod paused the unwatched catalog provider; approval waited 10 s and failed. Found while timing D-1. Fixed: read the repository once (PR #70).

DASHBOARD SPUN OFFLINE

`count()` waits for the server forever. Found while shooting the error state. Fixed: 15 s timeout → error state (PR #71).

ERROR STATES 3 MIN LATE

Riverpod 3 retried failed providers 10× while showing loading. Found on the deployed web admin. Fixed: `retry: noRetry` in both apps (PR #77).

ROLE READ ONLY ONCE

A demoted staff kept the admin UI until reload (CR-F1). Fixed: the session follows the user's role document live (PR #75).

GEMINI MODEL RETIRED

`gemini-2.5-flash` returned 404 for new projects. Fixed: default `gemini-flash-lite-latest`, model stays in Remote Config.

AI NEEDS A PLAN B

App Check, quota and model changes all broke AI at some point; the hand-written path kept the flow alive.